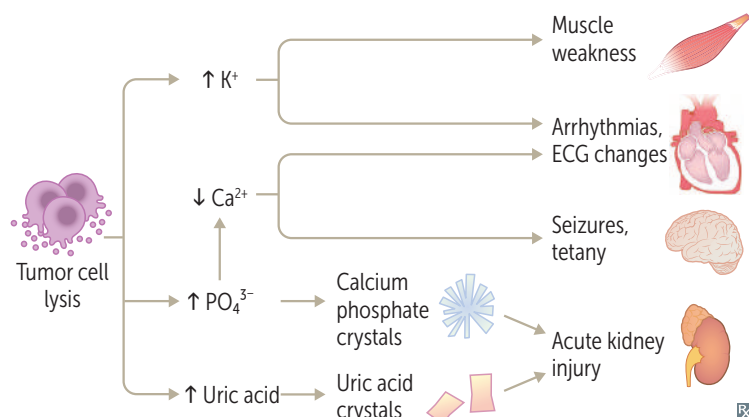


Tumor lysis syndrome

Oncologic emergency triggered by massive tumor cell lysis, seen most often with lymphomas/leukemias. Usually caused by treatment initiation, but can occur spontaneously with fast-growing cancers. Release of K^+ $\rightarrow$ hyperkalemia, release of PO_4^{3-} $\rightarrow$ hyperphosphatemia, hypocalcemia due to Ca^{2+} sequestration by PO_4^{3-} . $\uparrow$ nucleic acid breakdown $\rightarrow$ hyperuricemia $\rightarrow$ acute kidney injury. Prevention and treatment include aggressive hydration, allopurinol, rasburicase. Tumor cell breakdown causes $\uparrow K^+$, $\downarrow Ca^{2+}$, $\uparrow$ Uric acid, and $\uparrow PO_4^{3-}$ (Break the CUP).

► HEMATOLOGY AND ONCOLOGY—PHARMACOLOGY**Heparin****MECHANISM**

Activates antithrombin, which $\downarrow$ action primarily of factors IIa (thrombin) and Xa. Short half-life.

CLINICAL USE

Immediate anticoagulation for pulmonary embolism (PE), acute coronary syndrome, MI, deep venous thrombosis (DVT). Used during pregnancy (does not cross placenta; low-molecular-weight heparin [eg, enoxaparin] preferred). Monitor PTT.

ADVERSE EFFECTS

Bleeding (reverse with protamine sulfate), heparin-induced thrombocytopenia (HIT), osteoporosis (with long-term use), drug-drug interactions, type 4 renal tubular acidosis.

- **HIT type 1**—mild (platelets $> 100,000/mm^3$), transient, nonimmunologic drop in platelet count that typically occurs within the first 2 days of heparin administration. Not clinically significant.
- **HIT type 2**—development of IgG antibodies against heparin-bound platelet factor 4 (PF4) that typically occurs 5–10 days after heparin administration. Antibody-heparin-PF4 complex binds and activates platelets $\rightarrow$ removal by splenic macrophages and thrombosis $\rightarrow$ $\downarrow\downarrow$ platelet count. Highest risk with unfractionated heparin. Treatment: discontinue heparin, start alternative anticoagulant (eg, argatroban). Fondaparinux safe to use (does not interact with PF4).

NOTES

Low-molecular-weight heparins (eg, enoxaparin, dalteparin) act mainly on factor Xa. Longer half-life. Fondaparinux acts only on factor Xa. Both are not easily reversible. Unfractionated heparin used in patients with renal insufficiency (low-molecular-weight heparins should be used with caution because they undergo renal clearance).

Warfarin

MECHANISM	Inhibits vitamin K epoxide reductase by competing with vitamin K → inhibition of vitamin K–dependent γ -carboxylation of clotting factors II, VII, IX, and X and proteins C and S. Metabolism affected by polymorphisms in the gene for vitamin K epoxide reductase complex (VKORC1). In laboratory assay, has effect on extrinsic pathway and ↑ PT . Long half-life. “The ex-President went to war (farin).”
CLINICAL USE	Chronic anticoagulation (eg, venous thromboembolism prophylaxis and prevention of stroke in atrial fibrillation). Not used in pregnant patients (because warfarin, unlike heparin, crosses placenta). Monitor PT/INR.
ADVERSE EFFECTS	Bleeding, teratogenic effects, skin/tissue necrosis A , drug-drug interactions (metabolized by cytochrome P-450 [CYP2C9]). Initial risk of hypercoagulation: protein C has shorter half-life than factors II and X. Existing protein C depletes before existing factors II and X deplete, and before warfarin can reduce factors II and X production → hypercoagulation. Skin/tissue necrosis within first few days of large doses believed to be due to small vessel microthrombosis. Heparin “bridging”: heparin frequently used when starting warfarin. Heparin’s activation of antithrombin enables anticoagulation during initial, transient hypercoagulable state caused by warfarin. Initial heparin therapy reduces risk of recurrent venous thromboembolism and skin/tissue necrosis. For reversal of warfarin, give vitamin K. For rapid reversal, give FFP or PCC.

**Heparin vs warfarin**

	Heparin	Warfarin
ROUTE OF ADMINISTRATION	Parenteral (IV, SC)	Oral
SITE OF ACTION	Blood	Liver
ONSET OF ACTION	Rapid (seconds)	Slow, limited by half-lives of normal clotting factors
DURATION OF ACTION	Hours	Days
MONITORING	PTT (in trinsic pathway)	PT/INR (extrinsic pathway)
CROSSES PLACENTA	No	Yes (teratogenic)

Direct coagulation factor inhibitors Do not usually require lab monitoring.

DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Bivalirudin, argatroban, dabigatran	Directly inhibit thrombin (factor IIa)	Venous thromboembolism, atrial fibrillation. Can be used in HIT, when heparin is BAD for the patient	Bleeding (ida rucizumab can be used to inhibit dabi gatran)
Apixaban, edoxaban, rivaroxaban	Directly inhibit (ban) factor Xa	Oral agents. DVT/PE treatment and prophylaxis; stroke prophylaxis in patients with atrial fibrillation	Bleeding (reverse with andexa net alfa)

Anticoagulation reversal

ANTICOAGULANT	REVERSAL AGENT	NOTES
Heparin	Protamine sulfate	⊕ charged peptide that binds ⊖ charged heparin
Warfarin	Vitamin K (slow) +/- FFP or PCC (rapid)	
Dabigatran	Idarucizumab	Monoclonal antibody Fab fragments
Direct factor Xa inhibitors (eg, apixaban, rivaroxaban)	Andexanet alfa	Recombinant modified factor Xa (inactive)

Antiplatelets

All work by ↓ platelet aggregation.

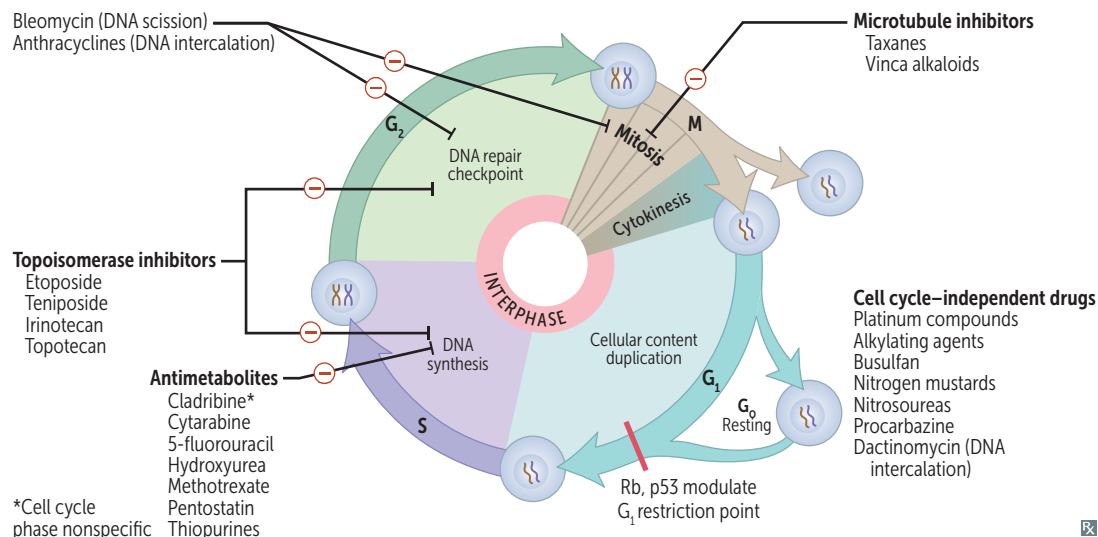
DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Aspirin	Irreversibly blocks COX → ↓ TXA ₂ release	Acute coronary syndrome; coronary stenting. ↓ incidence or recurrence of thrombotic stroke	Gastric ulcers, tinnitus, allergic reactions, renal injury, Reye syndrome (in children)
Clopidogrel, prasugrel, ticagrelor	Block ADP (P2Y ₁₂) receptor → ↓ ADP-induced expression of GpIIb/IIIa	Same as aspirin; dual antiplatelet therapy	Bleeding
Eptifibatide, tirofiban, abciximab	Block GpIIb/IIIa (fibrinogen receptor) on activated platelets	Unstable angina, percutaneous coronary intervention	Bleeding, thrombocytopenia
Cilostazol, dipyridamole	Block phosphodiesterase → ↓ cAMP hydrolysis → ↑ cAMP in platelets	Intermittent claudication, stroke prevention, cardiac stress testing, prevention of coronary stent restenosis	Nausea, headache, facial flushing, hypotension, abdominal pain

Thrombolytics

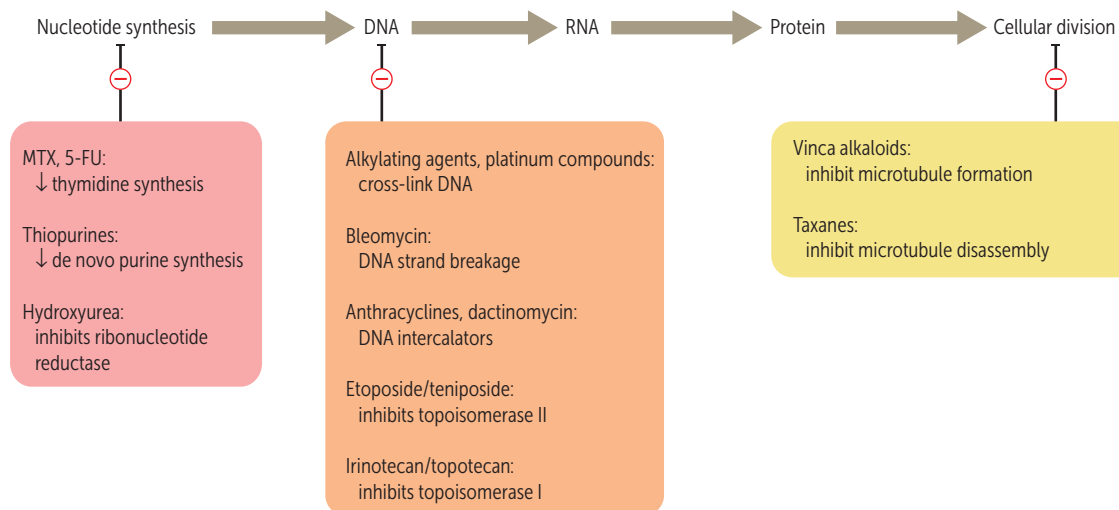
Alteplase (tPA), reteplase (rPA), tenecteplase (TNK-tPA).

MECHANISM	Directly or indirectly aid conversion of plasminogen to plasmin, which cleaves fibrin clots. ↑ PT, ↑ PTT, no change in platelet count.
CLINICAL USE	Early MI, early ischemic stroke, direct thrombolysis of high-risk PE.
ADVERSE EFFECTS	Bleeding. Contraindicated in patients with active bleeding, history of intracranial bleeding, recent surgery, known bleeding diatheses, or severe hypertension. Nonspecific reversal with antifibrinolytics (eg, aminocaproic acid, tranexamic acid), platelet transfusions, and factor corrections (eg, cryoprecipitate, FFP, PCC).

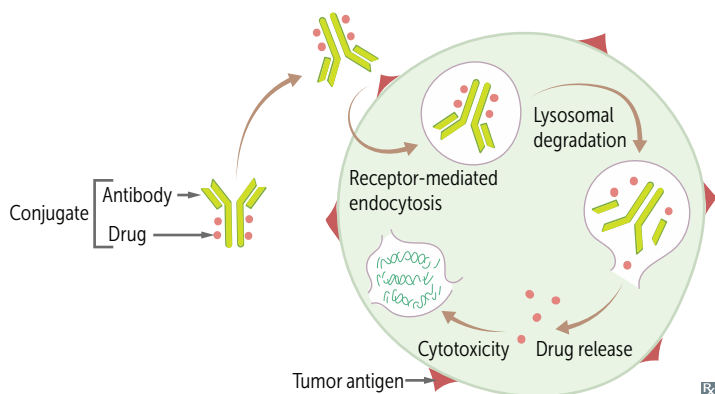
Cancer therapy—cell cycle



Cancer therapy—targets



Antibody-drug conjugates



Formed by linking monoclonal antibodies with cytotoxic chemotherapeutic drugs. Antibody selectivity against tumor antigens allows targeted drug delivery to tumor cells while sparing healthy cells → ↑ efficacy and ↓ toxicity. Example: ado-trastuzumab emtansine (T-DM1) for HER2 ⊕ breast cancer.

Antitumor antibiotics	Dactinomycin is cell cycle nonspecific; bleomycin and anthracycline are G ₂ /M phase specific.		
DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Bleomycin	Generates free radicals → DNA strand breaks	Testicular cancer, Hodgkin lymphoma	Pulmonary fibrosis, skin hyperpigmentation
Dactinomycin (actinomycin D)	Intercalates into DNA, preventing RNA synthesis	Wilms tumor, Ewing sarcoma, rhabdomyosarcoma	Myelosuppression
Anthracyclines Doxorubicin, daunorubicin	Generate free radicals Intercalate into DNA → DNA strand breaks → ↓ replication Inhibit topoisomerase II	Solid tumors, leukemias, lymphomas	Dilated cardiomyopathy (often irreversible; prevent with dexrazoxane), myelosuppression

Antimetabolites	All are S-phase specific except cladribine, which is cell cycle nonspecific.		
DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Thiopurines Azathioprine, 6-mercaptopurine	Purine (thiol) analogs → ↓ de novo purine synthesis AZA is converted to 6-MP, which is then activated by HGPRT	Rheumatoid arthritis, IBD, SLE, ALL; steroid-refractory disease Prevention of organ rejection Weaning from glucocorticoids	Myelosuppression; GI, liver toxicity 6-MP is inactivated by thiopurine S-methyltransferase (genetic polymorphism) and xanthine oxidase (↑ toxicity with allopurinol or febuxostat)
Cladribine, pentostatin	Purine nucleoside analogs → unable to be processed by ADA, interfering with DNA synthesis	Hairy cell leukemia	Myelosuppression
Cytarabine (arabinofuranosyl cytidine)	Pyrimidine nucleoside analog → DNA chain termination Inhibits DNA polymerase	Leukemias (AML), lymphomas	Myelosuppression
5-Fluorouracil	Pyrimidine analog bioactivated to 5-FdUMP → thymidylate synthase inhibition → ↓ dTMP → ↓ DNA synthesis Capecitabine is a prodrug	Colon cancer, pancreatic cancer, actinic keratosis, basal cell carcinoma (topical) Effects enhanced with the addition of leucovorin	Myelosuppression, palmar-plantar erythrodysesthesia (hand-foot syndrome)
Hydroxyurea	Inhibits ribonucleotide reductase → ↓ DNA synthesis	Myeloproliferative disorders (eg, CML, polycythemia vera), sickle cell disease (↑ HbF)	Severe myelosuppression, megaloblastic anemia
Methotrexate	Folic acid analog that competitively inhibits dihydrofolate reductase → ↓ dTMP → ↓ DNA synthesis	Cancers: leukemias (ALL), lymphomas, choriocarcinoma, sarcomas Nonneoplastic: ectopic pregnancy, medical abortion (with misoprostol), rheumatoid arthritis, psoriasis, IBD, vasculitis	Myelosuppression (reversible with leucovorin “rescue”), hepatotoxicity, mucositis (eg, mouth ulcers), pulmonary fibrosis, folate deficiency (teratogenic), nephrotoxicity

Alkylating agents All are cell cycle nonspecific.

DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Busulfan	Cross-links DNA	Used to ablate patient's bone marrow before bone marrow transplantation	Severe myelosuppression (in almost all cases), pulmonary fibrosis, hyperpigmentation
Nitrogen mustards Cyclophosphamide, ifosfamide	Cross-link DNA Require bioactivation by liver	Solid tumors, leukemia, lymphomas, rheumatic disease (eg, SLE, granulomatosis with polyangiitis)	Myelosuppression, SIADH, Fanconi syndrome (ifosfamide), hemorrhagic cystitis and bladder cancer (prevent with mesna)
Nitrosoureas Carmustine, lomustine	Cross-link DNA Require bioactivation by liver Cross blood-brain barrier → CNS entry	Brain tumors (including glioblastoma multiforme) Put nitro in your Mustang and travel the globe	CNS toxicity (convulsions, dizziness, ataxia)
Procarbazine	Mechanism unknown Weak MAO inhibitor (risk of hypertensive crisis with tyramine ingestion)	Hodgkin lymphoma, brain tumors	Myelosuppression, pulmonary toxicity, leukemia, disulfiram-like reaction

Platinum compounds Cisplatin, carboplatin, oxaliplatin.

MECHANISM	Cross-link DNA. Cell cycle nonspecific.
CLINICAL USE	Solid tumors (eg, testicular, bladder, ovarian, GI, lung), lymphomas.
ADVERSE EFFECTS	Nephrotoxicity (eg, Fanconi syndrome; prevent with amifostine), peripheral neuropathy, ototoxicity.

Microtubule inhibitors All are M-phase specific.

DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Taxanes Docetaxel, paclitaxel	Hyper stabilize polymerized microtubules → prevent mitotic spindle breakdown	Various tumors (eg, ovarian and breast carcinomas)	Myelosuppression, neuropathy, hypersensitivity Taxes stabilize society
Vinca alkaloids Vincristine, vinblastine	Bind β -tubulin and inhibit its polymerization into microtubules → prevent mitotic spindle formation	Solid tumors, leukemias, Hodgkin and non-Hodgkin lymphomas	Vin cr istine (crisps the nerves): neurotoxicity (axonal neuropathy), constipation (including ileus) Vin blast ine (blasts the marrow): myelosuppression

Topoisomerase inhibitors All cause ↑ DNA strand breaks, resulting in cell cycle arrest in S and G₂ phases.

DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Irinotecan, topotecan	Inhibit topoisomerase I “-te cone ”	Colon, ovarian, small cell lung cancer	Severe myelosuppression, diarrhea
Etoposide, teniposide	Inhibit topoisomerase II “-bo th side”	Testicular, small cell lung cancer, leukemia, lymphoma	Myelosuppression, alopecia

Tamoxifen

MECHANISM	Selective estrogen receptor modulator with complex mode of action: antagonist in breast tissue, partial agonist in endometrium and bone. Blocks the binding of estrogen to ER in ER ⊕ cells.
CLINICAL USE	Prevention and treatment of breast cancer, prevention of gynecomastia in patients undergoing prostate cancer therapy.
ADVERSE EFFECTS	Hot flashes, ↑ risk of thromboembolic events (eg, DVT, PE), endometrial cancer, uterine sarcoma.

Anticancer monoclonal antibodies Work against extracellular targets to neutralize them or to promote immune system recognition (eg, ADCC by NK cells). Eliminated by macrophages (not cleared by kidneys or liver).

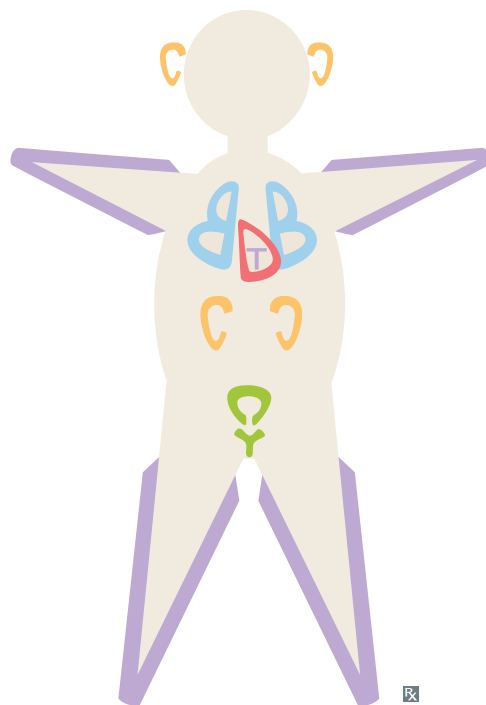
AGENT	TARGET	CLINICAL USE	ADVERSE EFFECTS
Alemtuzumab	CD52	Chronic ly mphocytic leukemia (CLL), multiple sclerosis.	↑ risk of infections and autoimmunity (eg, ITP)
Bevacizumab	VEGF (inhibits b lood v essel formation)	Colorectal cancer (CRC), renal cell carcinoma (RCC), non–small cell lung cancer (NSCLC), angioproliferative retinopathy	Hemorrhage, blood clots, impaired wound healing
Cetuximab, panitumumab	EGFR	Metastatic CRC (wild-type RAS), head and neck cancer	Rash, elevated LFTs, diarrhea
Rituximab	CD20	Non-Hodgkin lymphoma, CLL, rheumatoid arthritis, ITP, TTP, AIHA, multiple sclerosis	Infusion reaction due to cytokine release following interaction of rituximab with its target on B cells
Trastuzumab	HER2 Don't Trast HER , she will break your heart	Breast cancer, gastric cancer	Dilated cardiomyopathy (often reversible)
Pembrolizumab, nivolumab, cemiplimab	PD-1	Various tumors (eg, NSCLC, RCC, melanoma, urothelial carcinoma)	↑ risk of autoimmunity (eg, dermatitis, enterocolitis, hepatitis, pneumonitis, endocrinopathies)
Atezolizumab, durvalumab, avelumab	PD-L1		
Ipilimumab	CTLA-4		

Anticancer small molecule inhibitors

AGENT	TARGET	CLINICAL USE	ADVERSE EFFECTS
Alectinib , crizotinib	ALK	Non–small cell lung cancer	Edema, rash, diarrhea
Erlotinib , gefitinib , afatinib	EGFR	Non–small cell lung cancer	Rash, diarrhea
Imatinib , dasatinib , nilotinib	BCR-ABL (also other tyrosine kinases [eg, c-KIT])	CML, ALL, GISTs	Myelosuppression, ↑ LFTs, edema, myalgias
Ruxolitinib	JAK1/2	Polycythemia vera	Bruises, ↑ LFTs
Bortezomib , ixazomib , carfilzomib	Proteasome (induce arrest at G2-M phase via accumulation of abnormal proteins → apoptosis)	Multiple myeloma, mantle cell lymphoma	Peripheral neuropathy, herpes zoster reactivation (↓ T-cell activation → ↓ cell-mediated immunity)
Vemurafenib , encorafenib , dabrafenib	BRAF	Melanoma Often co-administered with MEK inhibitors (eg, trametinib)	Rash, fatigue, nausea, diarrhea
Palbociclib	Cyclin -dependent kinase 4/6 (induces arrest at G1-S phase → apoptosis)	Breast cancer	Myelosuppression, pneumonitis
Olaparib	Poly (ADP-ribose) polymerase (↓ DNA repair)	Breast, ovarian, pancreatic, and prostate cancers	Myelosuppression, edema, diarrhea

Chemotoxicity amelioration

DRUG	MECHANISM	CLINICAL USE
Amifostine	Free radical scavenger	Nephrotoxicity from platinum compounds
Dexrazoxane	Iron chelator	Cardiotoxicity from anthracyclines
Leucovorin (folinic acid)	Tetrahydrofolate precursor	Myelosuppression from methotrexate (leucovorin “rescue”); also enhances the effects of 5-FU
Mesna	Sulfhydryl compound that binds acrolein (toxic metabolite of cyclophosphamide/ifosfamide)	Hemorrhagic cystitis from cyclophosphamide/ifosfamide
Rasburicase	Recombinant uricase that catalyzes metabolism of uric acid to allantoin	Tumor lysis syndrome
Ondansetron , granisetron	5-HT ₃ receptor antagonists	Acute nausea and vomiting (usually within 1-2 hr after chemotherapy)
Prochlorperazine , metoclopramide	D ₂ receptor antagonists	
Aprepitant , fosaprepitant	NK ₁ receptor antagonists	Delayed nausea and vomiting (>24 hr after chemotherapy)
Filgrastim , sargramostim	Recombinant G(M)-CSF	Neutropenia
Epoetin alfa	Recombinant erythropoietin	Anemia

Key chemotoxicities

Cisplatin, Carboplatin → ototoxicity

Vincristine → peripheral neuropathy

Bleomycin, Busulfan → pulmonary fibrosis

Doxorubicin, Daunorubicin → cardiotoxicity

Trastuzumab → cardiotoxicity

Cisplatin, Carboplatin → nephrotoxicity

CYclophosphamide → hemorrhagic cystitis

Nonspecific common toxicities of nearly all cytotoxic chemotherapies include myelosuppression (neutropenia, anemia, thrombocytopenia), GI toxicity (nausea, vomiting, mucositis), alopecia.